DAY 9 (4)

Firewall Basics Theory (1 hour): Firewall types and use cases (hardware and software firewalls). Practical (1 hour): Configure basic rules in a software firewall (e.g., Windows Defender). Task: Document how to block and allow specific traffic using a firewall.

Q1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Answer:

Steps to Create an Outbound Rule to Block Spotify

1. Press **Windows + R**, type **wf.msc**, and press **Enter**.
2. Windows Defender Firewall with Advanced Security will open.
3. Click **Outbound Rules** in the left panel.
4. Click **New Rule** in the right panel.
5. Select **Program** and click **Next**.
6. Choose **This program path**.
7. Browse and select the Spotify executable file:

C:\Users\<Username>\AppData\Roaming\Spotify\Spotify.exe

8. Click **Next**.
9. Select **Block the connection** and click **Next**.
10. Check **Domain**, **Private**, and **public** profiles.
11. Click **Next**.
12. Enter Rule Name:

Block Spotify Traffic

13. Click **Finish**.

Result:

All outbound network traffic generated by the Spotify desktop application will be blocked by Windows Defender Firewall.

Q2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Answer:

Steps to Create an Inbound Rule for Google Chrome

1. Open **Windows Defender Firewall with Advanced Security** (wf.msc).
2. Select **Inbound Rules**.
3. Click **New Rule**.
4. Choose **Program** and click **Next**.
5. Select **This program path** and browse to:

C:\Program Files\Google\Chrome\Application\chrome.exe

6. Click **Next**.
7. Select **Allow the connection** and click **Next**.
8. Select all profiles:
    - Domain
    - Private
    - Public
9. Click **Next**.

10. Enter Rule Name:

Allow Chrome HTTPS

11. Click **Finish**.

Configure Port 443

1. Open the rule properties.
2. Go to **Protocols and Ports** tab.
3. Select **TCP**.
4. Under **Local Port**, select **Specific Ports**.
5. Enter:

443

6. Click **Apply** and **OK**.

Result:

Google Chrome is allowed to receive incoming HTTPS connections on TCP port 443.

Q3. Document, step-by-step, how you would block all traffic from a specific IP address (123.45.67.89) using Windows Defender Firewall.

Answer:

Steps to Block Traffic from IP Address 123.45.67.89

1. Open **Windows Defender Firewall with Advanced Security** by typing:

wf.msc

2. Select **Inbound Rules**.
3. Click **New Rule**.
4. Select **Custom Rule** and click **Next**.
5. Leave **All Programs** selected.
6. Click **Next**.
7. Leave Protocol Type as **Any**.
8. Click **Next**.

Using the Scope Tab

9. Under **Remote IP Address**, select:

These IP Addresses

10. Click **Add**.
11. Enter:

123.45.67.89

12. Click **OK**.

Block the Connection

13. Click **Next**.
14. Select:

Block the Connection

15. Click **Next**.
16. Select all profiles:

- Domain
- Private
- Public

17. Click **Next**.
18. Enter Rule Name:

Block IP 123.45.67.89

19. Click **Finish**.

Result:

All inbound traffic originating from IP address **123.45.67.89** will be blocked by Windows Defender Firewall.

Q4. Compare Hardware Firewalls and Software Firewalls by listing 3 key differences and a use case for each.

Answer:

Feature	Hardware Firewall	Software Firewall
Installation	Installed as a dedicated physical device between network and Internet.	Installed directly on a computer or server.
Protection Scope	Protects the entire network.	Protects only the device on which it is installed.
Performance Impact	Minimal impact on individual computers.	Uses system resources such as CPU and RAM.
Cost	More expensive.	Usually free or low-cost.
Management	Centralized management for all devices.	Managed separately on each device.

Use Cases

Hardware Firewall

Example: Corporate office network.

Use Case:

A large company installs a hardware firewall between its internal network and the Internet to protect hundreds of computers from external threats.

Software Firewall

Example: Windows Defender Firewall.

Use Case:

A home user installs Windows Defender Firewall on a laptop to monitor and control incoming and outgoing network traffic.